QUESTION BANK

Unit-I: Cyber Security & Cryptography Basics

Easy

1. Explain the fundamental "Need for Security" in an increasingly digital society.
2. Define Plaintext and Ciphertext with a simple example of each.
3. Describe how a Caesar Cipher (a basic Mono-alphabetic Cipher) functions.
4. Illustrate the difference between a Substitution Cipher and a Transposition Cipher.
5. Classify the following as either Substitution or Transposition: Rail Fence, Playfair, Caesar.
6. Use the Rail Fence technique to encrypt the word "CYBERSECURITY" with 2 rails.
7. Explain the concept of "Polygram Substitution" and how it differs from mono-alphabetic methods.
8. Demonstrate how to decrypt a simple message using a keyword-based substitution cipher.
9. Describe the "Avalanche Effect" in the context of cryptographic strength.
10. Explain the role of a "Key" in ensuring that even if an algorithm is known, the message remains secure.

Moderate

11. Analyze why Mono-alphabetic ciphers are vulnerable to "Frequency Analysis."
12. Compare the security levels of Mono-alphabetic vs. Polyalphabetic substitution (like the Vigenere Cipher).
13. Examine how the Playfair Cipher handles double letters in a plaintext pair.
14. Determine the ciphertext for "MEET AT NOON" using a Columnar Transposition cipher with a key of "312".
15. Differentiate between "Diffusion" and "Confusion" as defined by Claude Shannon.
16. Calculate the total possible keys for a 5x5 Playfair Cipher matrix.
17. Implement a Vigenere Cipher manually for a short sentence given a 4-letter keyword.
18. Solve a cryptogram that has been encrypted using a simple Shift Cipher.

19. Determine why Transposition ciphers are often combined with Substitution ciphers in modern algorithms.
20. Examine the impact of "Polygram" techniques on making frequency analysis more difficult.

Hard

21. Critique the security of the Caesar Cipher—is it ever useful in the 21st century?
22. Evaluate the vulnerability of a Polyalphabetic cipher if the keyword length is known.
23. Design a custom "Double Transposition" scheme and explain why it is harder to break than a single transposition.
24. Justify the shift from manual classical ciphers to machine-based cryptography during WWII.
25. Appraise the risk of using a "Short Key" in a Vigenere cipher against Kasiski Examination.
26. Construct a 5x5 matrix for the Playfair cipher using the keyword "SECURITY" and encrypt "HELLO."
27. Formulate a strategy for a "Chosen Plaintext Attack" on a simple substitution cipher.
28. Argue why "Security through Obscurity" (hiding the algorithm) is considered a bad practice in modern cryptography.
29. Propose a method to identify if a ciphertext was created using a Transposition or a Substitution cipher.
30. Develop a Python script logic to perform a brute-force attack on a Caesar Cipher.

Unit-II: Encryption, Symmetric Key & AES/RSA

Easy

1. Explain the core difference between Symmetric and Asymmetric Key Cryptography.
2. Describe the "Key Distribution Problem" in symmetric cryptography.
3. Illustrate the high-level steps of the RSA algorithm (Key Generation, Encryption, Decryption).
4. Use the concept of "Prime Factorization" to explain why RSA is considered secure.
5. Classify DES and AES as Symmetric Key Block Ciphers.

6. Demonstrate how a sender uses a recipient's Public Key to send a private message.
7. Explain the "Rounds" in DES—why does the algorithm repeat the process 16 times?
8. Show the basic structure of a Feistel Network used in DES.
9. Identify the bit-sizes for the key and block in the standard DES algorithm.
10. Describe the "Confusion" and "Diffusion" properties provided by the S-Boxes in DES.

Moderate

11. Analyze why AES (Advanced Encryption Standard) replaced DES as the global standard.
12. Compare RSA and DES in terms of computational speed and resource usage.
13. Examine how the "Trapdoor One-Way Function" is used in the RSA algorithm.
14. Determine the public and private keys in RSA if $p=3$ and $q=11$.
15. Differentiate between "Block Ciphers" and "Stream Ciphers."
16. Calculate the effective key length of DES and explain why it is 56 bits instead of 64.
17. Implement the "SubBytes" and "ShiftRows" steps of the AES algorithm conceptually.
18. Solve an RSA encryption for message $M=2$ given public key $(e=7, n=33)$.
19. Determine why a 1024-bit RSA key is becoming "weak" by modern security standards.
20. Examine the "Meet-in-the-middle" attack on Double-DES and how Triple-DES (3DES) mitigates it.

Hard

21. Critique the use of DES in modern banking—should it be entirely phased out for AES?
22. Evaluate the security of RSA against the potential of "Quantum Computing."
23. Design a hybrid system that uses RSA to exchange a key and AES to encrypt the actual data.
24. Justify the use of S-Boxes in DES—what happens to the security if they are removed?
25. Appraise the trade-offs between key length and system performance in mobile devices.

26. Construct a step-by-step mathematical proof of how RSA decryption retrieves the original message.
27. Formulate a defense against "Brute Force" attacks for symmetric keys.
28. Argue why "Public Key Infrastructure (PKI)" is necessary for secure internet communication.
29. Propose a scenario where Symmetric encryption is mandatory and Asymmetric is not feasible.
30. Develop a logic to detect a "Weak Key" in a symmetric algorithm.

Unit-III: Network Security, Firewalls & VPNs

Easy

1. Explain the 4-layer architecture of the TCP/IP model.
2. Describe the role of a "Firewall" in a corporate network.
3. Illustrate how a "Virtual Private Network" (VPN) creates a secure tunnel over the internet.
4. Use the "Three-Way Handshake" to explain how a TCP connection is established.
5. Classify attacks into "Passive Attacks" (Eavesdropping) and "Active Attacks" (Modification).
6. Demonstrate how SHTTP (Secure HTTP) differs from standard HTTP in terms of encryption.
7. Explain the purpose of a "Time Stamping Protocol" (TSP) in digital transactions.
8. Describe a "Denial of Service" (DoS) attack in simple terms.
9. Identify the components involved in a "Secure Electronic Transaction" (SET).
10. Explain what a "Proxy Server" does in the context of network security.

Moderate

11. Analyze the difference between a "Packet Filtering Firewall" and an "Application-level Gateway."
12. Compare the security of "Tunnel Mode" vs "Transport Mode" in IPsec VPNs.
13. Examine how a "Man-in-the-Middle" (MITM) attack can intercept encrypted traffic.
14. Determine the best type of firewall for a company that wants to block specific web applications (like social media).

15. Differentiate between "Intrusion Detection Systems" (IDS) and "Intrusion Prevention Systems" (IPS).
16. Calculate the potential impact of a "Distributed DoS" (DDoS) attack using a Botnet.
17. Implement a basic security rule for a firewall to block all incoming traffic except on Port 80 and 443.
18. Solve a connectivity issue where a VPN is blocked by a local network firewall.
19. Determine why "Digital Certificates" are essential for SET to verify the identity of the merchant and buyer.
20. Examine the vulnerabilities of the TCP/IP stack (e.g., IP Spoofing, SYN Flooding).

Hard

21. Critique the "Zero Trust Architecture" vs. traditional "Perimeter-based Security" (Firewalls).
22. Evaluate the effectiveness of SHTTP for protecting high-value financial data.
23. Design a network topology for a bank that includes DMZs, Firewalls, and VPN gateways.
24. Justify the use of "Honeypots" as a proactive network security measure.
25. Appraise the legal and privacy concerns of "Deep Packet Inspection" (DPI) by ISPs.
26. Construct a protocol for "Secure Electronic Transaction" that prevents a merchant from seeing a customer's credit card number.
27. Formulate a response plan for a "Ransomware" attack that has bypassed the network firewall.
28. Argue why "Encryption at Rest" and "Encryption in Transit" must both be implemented.
29. Propose a multi-layered defense strategy against "SQL Injection" and "Cross-Site Scripting" (XSS).
30. Develop a checklist for auditing a company's "Virtual Private Network" security.

Unit-IV: Information Systems & Security Risk Analysis

Easy

1. Explain what an "Information System" is and name three different types (e.g., MIS, DSS).
2. Describe the "Information Life Cycle" (Creation, Storage, Usage, Destruction).
3. Illustrate the "Need for Information Security" using the CIA Triad.
4. Use an example to define a "Threat" versus a "Vulnerability."
5. Classify common threats into "Natural Threats" (Floods) and "Human-made Threats" (Hacking).
6. Demonstrate how "Security Risk Analysis" helps in prioritizing security budgets.
7. Explain the concept of "Information Assurance" (IA) and its five pillars.
8. Describe the impact of a data breach on a company's brand reputation.
9. Identify internal threats (disgruntled employees) vs external threats (organized cybercrime).
10. Explain why "Risk Acceptance" is sometimes a valid strategy.

Moderate

11. Analyze the stages of "Risk Management": Identification, Assessment, Mitigation, and Monitoring.
12. Compare Qualitative Risk Analysis (High/Medium/Low) with Quantitative Risk Analysis (Monetary Value).
13. Examine how "Shadow IT" (employees using unapproved apps) creates new security risks.
14. Determine the "Annualized Loss Expectancy" (ALE) for a server if the Single Loss Expectancy (SLE) is $5,000 and it happens twice a year.
15. Differentiate between "Risk Mitigation" and "Risk Transference" (e.g., Cyber Insurance).
16. Calculate the risk score for a system with a high impact but low probability of attack.
17. Implement a "Least Privilege" access policy for a new department's information system.
18. Solve a scenario where a system's "Availability" is compromised by an outdated hardware component.
19. Determine the critical assets in a hospital's information system that require the highest level of assurance.
20. Examine the "Social Engineering" threats to information systems (Phishing, Tailgating).

Hard

21. Critique the "BYOD" (Bring Your Own Device) policy from a security risk perspective.
22. Evaluate the "Cost-Benefit Analysis" of implementing a multi-million dollar security system for a small business.
23. Design an "Information Security Awareness" program for non-technical employees.
24. Justify why "Integrity" is more important than "Confidentiality" for a medical record system.
25. Appraise the risk of "Cloud Migration" for a government agency dealing with sensitive citizen data.
26. Construct a "Business Impact Analysis" (BIA) report for a major database failure.
27. Formulate a "Disaster Recovery Plan" (DRP) that includes off-site backups and warm sites.
28. Argue for the necessity of "Continuous Monitoring" instead of annual security audits.
29. Propose a method for measuring the "Security ROI" (Return on Investment).
30. Develop a "Risk Register" for an e-commerce website that lists five potential threats and their mitigations.

Unit-V: Security Policies & Legal Frameworks

Easy

1. Explain why a company needs a written "Security Policy."
2. Describe the "Acceptable Use Policy" (AUP) for company email.
3. Illustrate the importance of the "ISO 27001" standard in information security.
4. Use the "IT Act 2000" (India) to explain the legal recognition of digital signatures.
5. Classify intellectual property into Copyrights, Patents, and Trademarks.
6. Demonstrate the requirement for "Publishing and Notification" when a new security policy is enacted.
7. Explain the "Right to Privacy" as it relates to corporate data collection.
8. Describe what a "WWW Policy" should cover regarding employee web browsing.
9. Identify the main goals of the "Copyright Act" in the digital age.
10. Explain the role of a "Certifying Authority" under the IT Act.

Moderate

11. Analyze the 14 domains of the "ISO/IEC 27001" framework.
12. Compare the "Patent Law" protections for hardware vs software algorithms.
13. Examine the penalties for "Hacking" and "Data Theft" under the Indian IT Act.
14. Determine the necessary components of a "Corporate Password Policy."
15. Differentiate between "Legal Compliance" and "Security Best Practices."
16. Calculate the potential fines a company might face for failing to protect customer data under the IT Act.
17. Implement a "Non-Disclosure Agreement" (NDA) clause in a vendor security policy.
18. Solve a conflict where an employee's "Right to Privacy" clashes with a company's "Email Monitoring Policy."
19. Determine how "Fair Use" applies to copyrighted material found on the internet.
20. Examine the "Safe Harbor" provisions for internet intermediaries (like ISPs) under the IT Act.

Hard

21. Critique the effectiveness of the "IT Act 2000" in dealing with modern social media crimes.
22. Evaluate the impact of "GDPR" (General Data Protection Regulation) on global security policies.
23. Design a "Security Policy" for a startup that allows employees to work remotely.
24. Justify the need for "Mandatory Breach Notification" policies in all corporate sectors.
25. Appraise the ethics of "Software Patenting"—does it help or hinder innovation?
26. Construct a "Corporate Incident Response Policy" that outlines who to contact during a breach.
27. Formulate a plan for "Policy Compliance Auditing"—how do you ensure employees are following the rules?
28. Argue whether "Privacy" is an absolute right in the context of "National Security" laws.
29. Propose a revision to the "IT Act" to better handle "Deepfakes" and "AI-generated Fraud."
30. Develop a "Social Media Policy" for employees to prevent the accidental leak of corporate secrets.